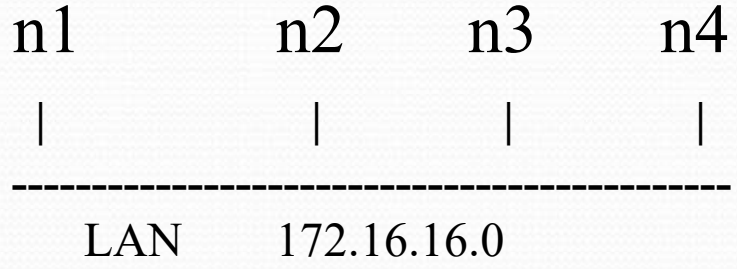
Simulation of Wired Networks in NS3

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Wired network Simulation

- Simulation of the Echo application for a simple wired network with four nodes connected in Bus topology



- New classes required for running this simulation:
  - CsmaHelper: used to manage the nodes for building a network with bus topology

CsmaHelper class

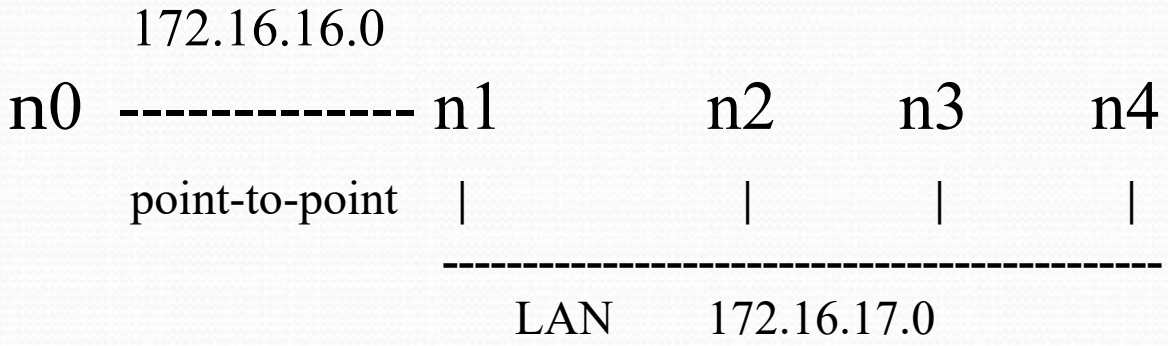
- CsmaHelper: used to manage the nodes for building a BUS network topology
  - NetDeviceContainer Install (Ptr <Node> node): It adds the device to the node and attaches the channel to the device.
  - NetDeviceContainer Install (const NodeContainer &c): install the device to all the nodes & attaches the channel to each devices.
  - SetChannelAttribute (std::string n1, const AttributeValue &v1): n1 & v1 are name and value of channel attribute.
  - SetDeviceAttribute (std::string n1, const AttributeValue &v1)

Simulation for Bus Network

- Create 4 nodes.
- Create the CSMA Channel with its attributes to create a LAN network topology.
- Install NetDevices in each nodes in the network.
- Install the InternetStack on each nodes.
- Assign the IP addresses to each devices in the network.
- Create the UdpEchoServer/Client applications.
- Install the application in corresponding nodes. (Eg., node 0 as server and other nodes as clients)
- Create animation interface and enable trace file.

Wired network Simulation

- Simulation for the Echo application for the following wired network; which has two nodes connected by P2P link and Four nodes connected in Bus topology.



Wired Network Simulation (Contd..)

- Node n0 and n1 are connected with p2p link & Node n1, n2, n3 & n4 are connected with CSMA link to form LAN network are the previous scenario that we discussed.

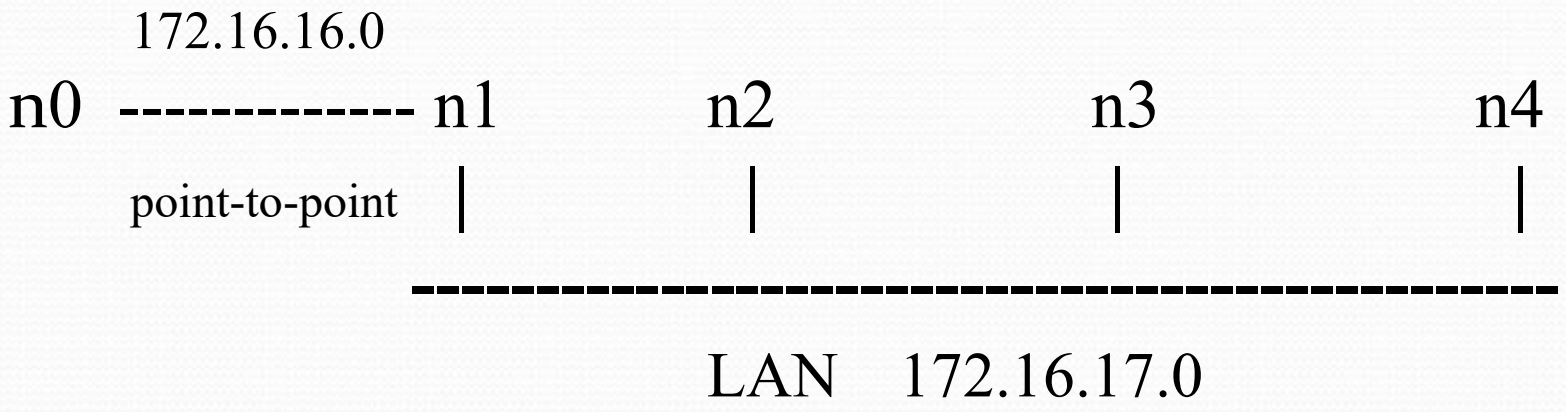
- Class required for running this simulation:
  - Ipv4GlobalRoutingHelper: use to perform simple routing to send packets to another network

Ipv4GlobalRoutingHelper Class

- Ipv4GlobalRoutingHelper: use to perform simple routing to send packets to another network
  - PopulateRoutingTables (void): Build a routing database and initialize the routing tables of the nodes in the simulation.
  - RecomputeRoutingTables (void): Remove all routes that were previously installed in a prior call to either PopulateRoutingTables() or RecomputeRoutingTables(), and add a new set of routes.
  - Create (Ptr<Node> node) : Creates the routing protocol on given node

Wired Network Simulation (Contd..)

- Simulate the Echo application from node n0 to n4 & analyze the performance.



- New steps:
  - Add 2 NetDevices for the node n1. One with p2p link and the other with csma link.
  - Use Ipv4GlobalRoutingHelper to perform simple routing.

Wired Network Simulation (Contd..)

- Suppose if the user wants to simulate with more packets/traffic in the network, then ns3 provides the following application classes:
  - BulkSendApplication class / its Helper class
    - Send more traffic as much as possible to fill the bandwidth up to MaxBytes or until the application is stopped (if MaxBytes is zero)
    - Only TCP sockets can be used, but UDP sockets can not be used. (Only SOCK_STREAM and SOCK_SEQPACKET sockets are supported)
  - PacketSinkApplication class / its Helper class
    - Used to receive and consume traffic generated to an IP address and port.

BulkSendHelper class

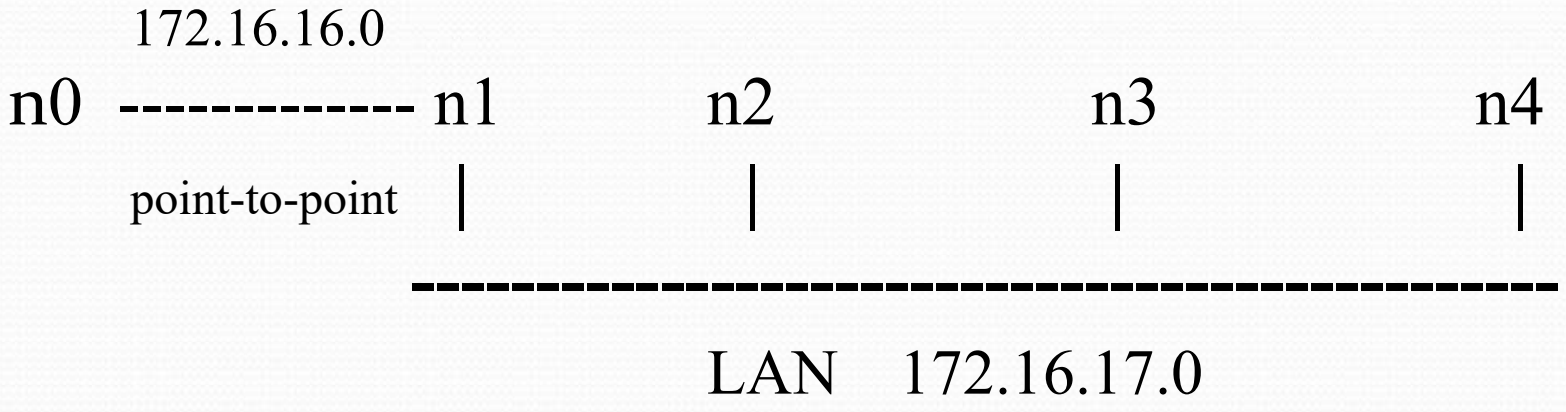
- Used to instantiate ns3::BulkSendApplication on a set of nodes.
- Data member:
  - m_factory : object factory (Eg., ns3::TcpSocketFactory)
- Constructor:
  - BulkSendHelper (std::string protocol, Address address)
    - protocol – name of the protocol used to send traffic by the application
    - address – address of the remote node to send the traffic
- Member functions:
  - ApplicationContainer Install (NodeContainer c): Install an ns3::BulkSendApplication on each node of the input container.
  - ApplicationContainer Install (Ptr<Node> node): Install an ns3::BulkSendApplication on the input node.
  - SetAttribute (std::string name, const AttributeValue &value)

PacketSinkHelper class

- Used to instantiate ns3::PacketSink on a set of nodes.
- Data member:
  - m_factory : object factory (Eg., ns3::TcpSocketFactory)
- Member functions:
  - ApplicationContainer Install (NodeContainer c): Install an ns3::PacketSink on each node of the input container.
  - ApplicationContainer Install (Ptr<Node> node): Install an ns3::PacketSink on the input node.
  - SetAttribute (std::string name, const AttributeValue &value)

Hands-on…

- Simulate the following network which can send/receive more number of packets from node n2, n3 and n4 to node n0 & analyze the network performance.



THANK YOU